

Aleksander Młynarski

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Profile

2nd-year Automation and Robotics student at AGH University and software developer at the AGH Solar Boat student research group. I develop software for autonomous systems in C++ and Python, working with ROS 2 in a Linux/Docker environment. Two-time finalist of the Physics Olympiad for the AGH Diamond Index. I am looking for an internship in software development or robotics.

Skills

Programming Languages: Python, C++

Systems & Architecture: ROS 2, Linux (WSL)

IT Tools & Simulations: Git, Docker, GitLab, MATLAB / Simulink, Autodesk Inventor

Experience

Software Engineer

10.2025 – present

AGH Solar Boat Student Research Group

- Designing and implementing Behavior Trees for decision-making logic in ROS 2.
- Writing optimal, rigorously tested C++ and Python code in compliance with industry standards.
- Containerizing the runtime environment using Docker tools.
- Managing code, resolving conflicts, and maintaining clean architecture in a GitLab repository.

Physics and Mathematics Tutor

03.2025 – present

Private Tutoring

- Conducting individual classes (15h/week) for high school and university students.
- Clearly explaining complex physics and mathematics concepts in an accessible manner.

Waiter

07.2024 – 08.2024

Restaurant in Kazimierz, Kraków

Selected Projects

- **Network Simulator (Netsim):** C++ implementation of a discrete-event simulator featuring graph-based topology modeling and input data integrity verification mechanisms.
- **Digital Signal Processing:** Implementation of the Discrete Fourier Transform (DFT) in MATLAB with matrix optimization, spectral analysis of signals, and spectral leakage reduction (Hamming and Gauss windows).
- **Numerical Simulations of a Robotic Gripper:** Development of analytical models of the mechanism, kinematic simulations, and finite element analysis (FEA) in Autodesk Inventor.

Education

B.Eng. in Automation Engineering and Robotics

10.2024 – present

AGH University of Science and Technology in Kraków

Mathematics, Physics & IT Profile

2020 – 2024

King John III Sobieski High School (No. 2) in Kraków

Languages

- **Polish** — Native
- **English** — C1
- **German** — C1

Certifications

- | | |
|---|---------|
| - Deutsches Sprachdiplom II (C1) | 02.2024 |
| - Deutsches Sprachdiplom I (B1) | 12.2020 |

Achievements & Interests

- **Achievements:** Two-time finalist of the Physics Olympiad for the AGH Diamond Index.
- **Interests:** Sports (swimming, gym, running), playing the guitar, reading.